PARISHRAM 2.0

CLASS 12TH 2026

HUMAN REPRODUCTION

ZOOLOGY | LECTURE-5

-PARAM SIR



TODAY'S TOPIC -

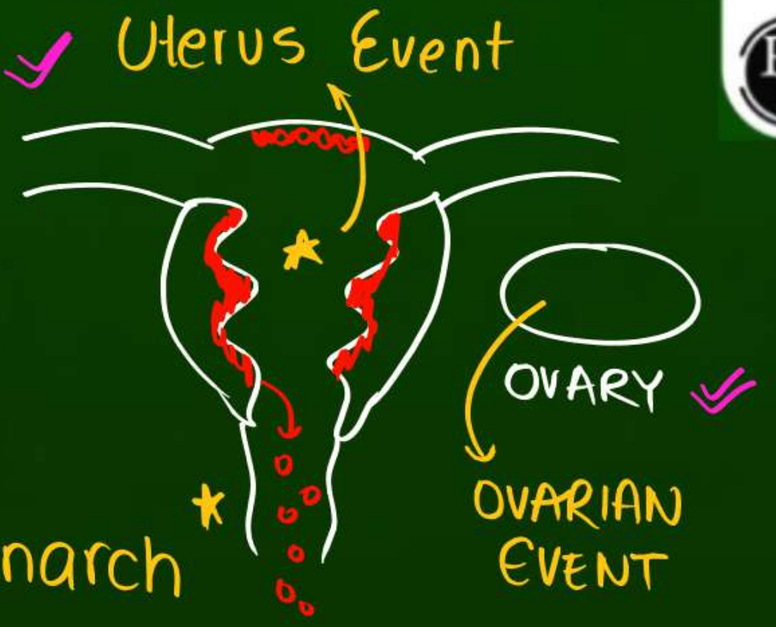
1. REVISION
2. MENSTRUAL CYCLE



Menstrual Cycle

1. Female primate - Menstrual cycle
2. Non primate - estrous cycle
3. When first menstrual cycle start : Menarch
4. When menstrual cycle Stops : Menopause
5. Duration 28/30 days
6. Start at puberty → ~ 50 years*

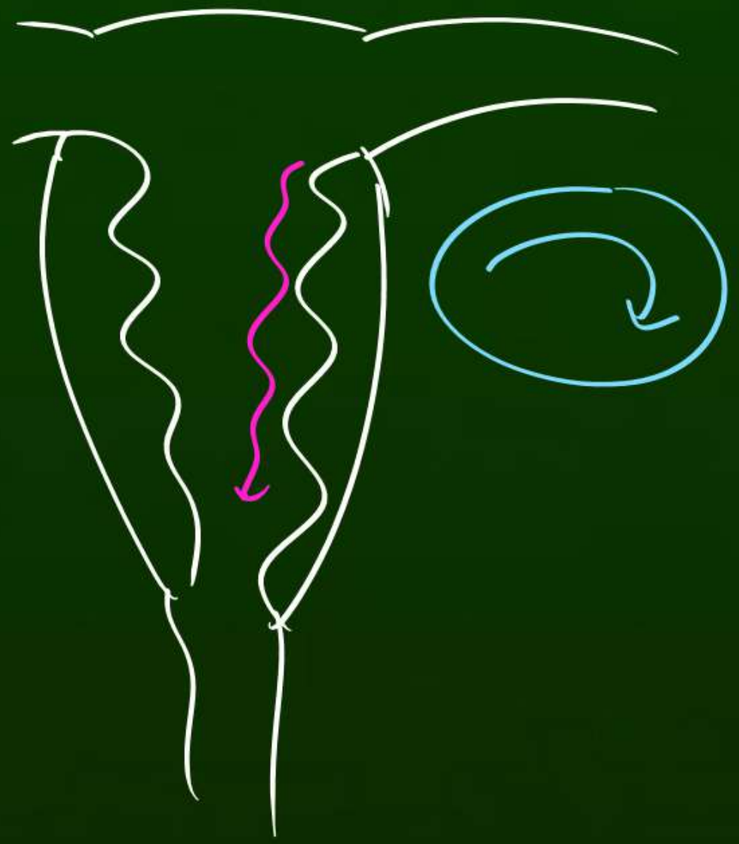
13 year





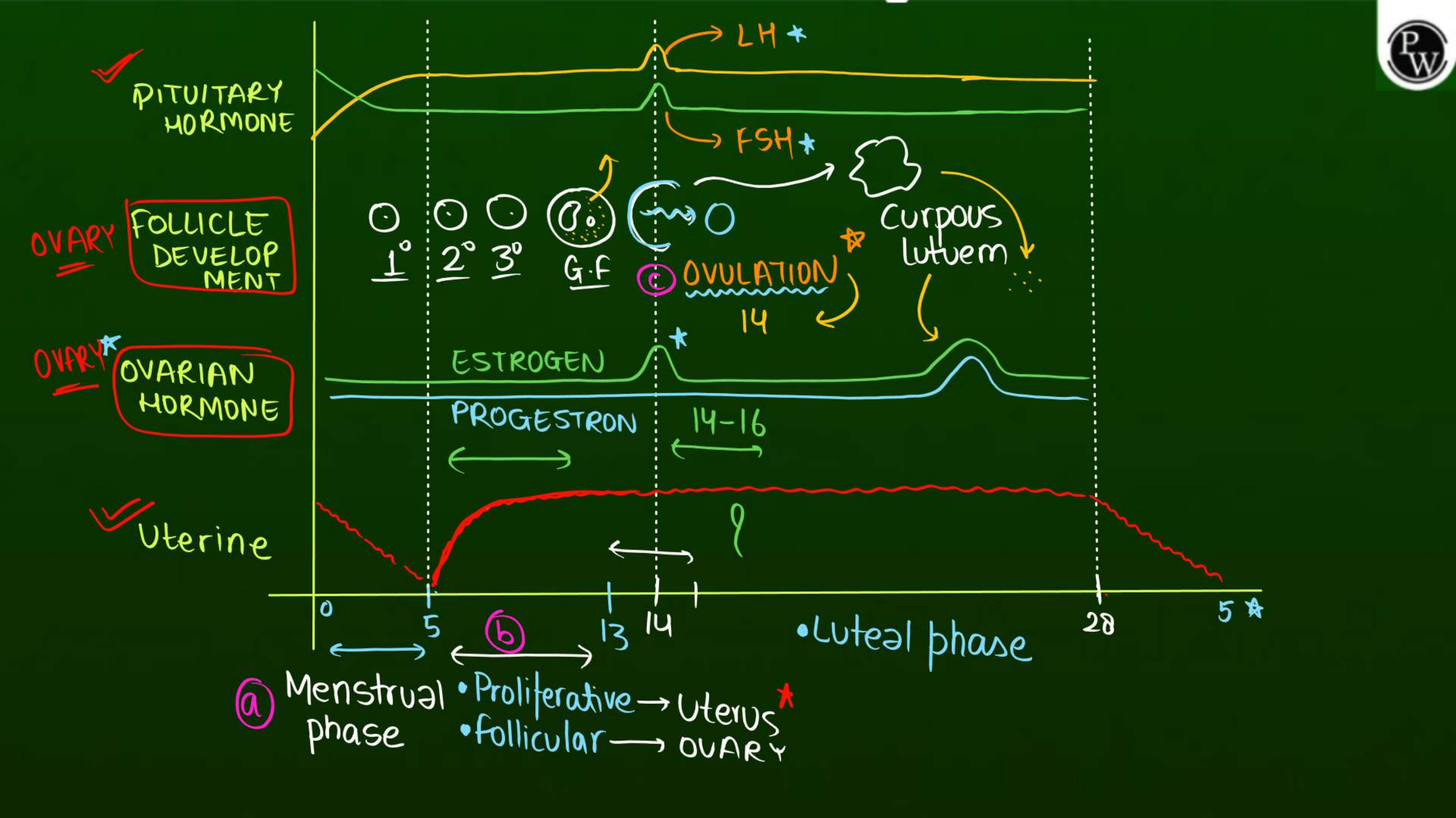
REVISION





Events | Phase in Menstrual Cycle

Uterus	OVARY
✓ ① Menstrual phase (1-5)	① Follicular phase
② Proliferative phase	② Ovulation phase
③ Secretory phase	③ Luteal phase


PITUITARY HORMONE
LH *
FSH *
OVARY
FOLLICLE DEVELOP MENT
1° 2° 3° G.F
OVULATION
14
Curpous luteum
OVARY*
OVARIAN HORMONE
ESTROGEN
PROGESTRON
14-16
Uterine
0 5 13 14 28 5 *
Luteal phase
@ Menstrual phase
b
• Proliferative → Uterus *
• follicular → OVARY

NCERT

EXTRACT

LH Surge

29
30
2

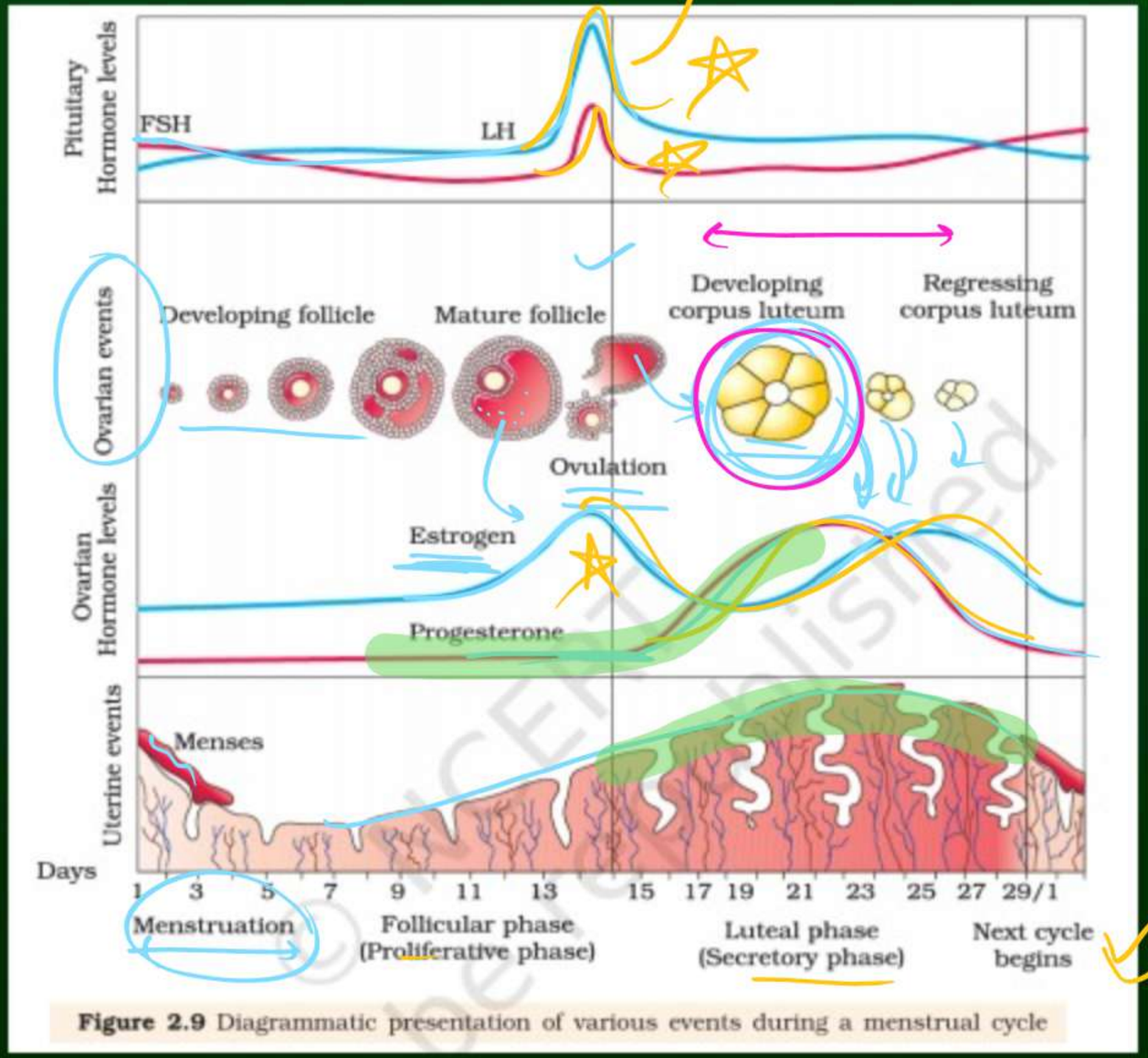


Figure 2.9 Diagrammatic presentation of various events during a menstrual cycle

hCG

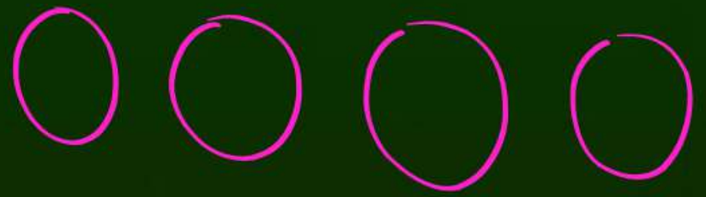
Placenta

C.L

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hCG : (human Chorionic Gonadotropin)

- Release during Pregnency by placenta
- act like LH
- help in maintain of Corpus luteum.



Menstrual hygine

→ Bath daily

→ Use Sanitary pad *

→ Change sanitary pad after 4-5 Hr

→ do't through in open.

→ Wash hand with Soap

Menstrual cup



2.4 Menstrual Cycle

The reproductive cycle in the female primates (e.g. monkeys, apes and human beings) is called menstrual cycle. The first menstruation begins at puberty and is called menarche. In human females, menstruation is repeated at an average interval of about 28/29 days, and the cycle of events starting from one menstruation till the next one is called the menstrual cycle. One ovum is released (ovulation) during the middle of each menstrual cycle. The major events of the menstrual cycle are shown in Figure 2.9. The cycle starts with the menstrual phase, when menstrual flow occurs and it lasts for 3-5 days. The menstrual flow results due to breakdown of endometrial lining of the uterus and its blood vessels which forms liquid that comes out through vagina. Menstruation only occurs if the released ovum is not fertilised. Lack of menstruation may be indicative of pregnancy. However, it may also be caused due to some other underlying causes like stress, poor health etc. The menstrual phase is followed by the follicular phase. During this phase, the primary follicles in the ovary grow to become a fully mature Graafian follicle and simultaneously the endometrium of uterus regenerates through proliferation. These changes in the ovary and the uterus are induced by changes in the levels of pituitary and ovarian hormones (Figure 2.9). The secretion of

gonadotropins (LH and FSH) increases gradually during the follicular phase, and stimulates follicular development as well as secretion of estrogens by the growing follicles. Both LH and FSH attain a peak level in the middle of cycle (about 14th day). Rapid secretion of LH leading to its maximum level during the mid-cycle called LH surge induces rupture of Graafian follicle and thereby the release of ovum (**ovulation**). The ovulation (ovulatory phase) is followed by the luteal phase during which the remaining parts of the Graafian follicle transform as the **corpus luteum** (Figure 2.9). The corpus luteum secretes large amounts of progesterone which is essential for maintenance of the endometrium. Such an endometrium is necessary for implantation of the fertilised ovum and other events of pregnancy. During pregnancy all events of the menstrual cycle stop and there is no menstruation. In the absence of fertilisation, the corpus luteum degenerates. This causes disintegration of the endometrium leading to menstruation, marking a new cycle. In human beings, menstrual cycles ceases around 50 years of age; that is termed as **menopause**. Cyclic menstruation is an indicator of normal reproductive phase and extends between menarche and menopause.

Menstrual Hygiene

Maintenance of hygiene and sanitation during menstruation is very important. Take bath and clean yourself regulary. Use sanitary napkins or clean homemade pads. Change sanitary napkins or homemade pads after every 4–5 hrs as per the requirement. Dispose of the used sanitary napkins properly wrapping it with a used paper. Do not throw the used napkins in the drainpipe of toilets or in the open area. After handling the napkin wash hands with soap.

QUESTION

'Menarche' term is used for

1. Cessation of menstrual cycle
2. First menstruation
3. Period of pregnancy in which menstruation ceases
4. Period of ovulation



QUESTION

The menstrual phase of menstrual cycle lasts for:

1. 8-15 days
2. 1-2 days
3. 1 day
4. 3-5 days



QUESTION

BIO

Phases of menstrual phase are followed as



1. Follicular phase, Menstrual phase and Luteal phase
2. Menstrual phase, follicular phase, proliferative phase
3. Menstrual phase, luteal phase and proliferative phase
4. Menstrual phase, proliferative phase and luteal phase

THANK YOU